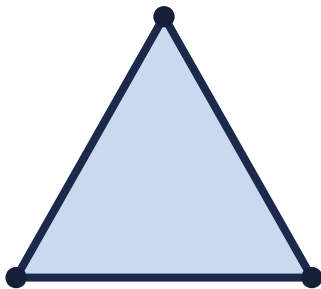


a cycle, not a boundary

counts as a hole

$$\partial c = 0, \quad c \notin \text{im } \partial$$



a cycle that IS a boundary

does not count

$$c = \partial(\text{triangle})$$